Quantification: 872_wt_hp_z04
==============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
---------

| measure | value |
|---|---|
| lacuna count | 21 |
| interior lacuna count | 18 |
| frame-edge lacuna count | 3 |

Interior lacunae only: mean, median and sample SD (n = 18)
----------------------------------------------------------

| measure | kind | mean | median | sd | units |
|---|---|---|---|---|---|
| area_px2 | lacuna shape | 1887.94 | 1578 | 1120.7 | px^2 |
| major_axis_length_px | lacuna shape | 93.3239 | 76.683 | 59.7942 | px |
| minor_axis_length_px | lacuna shape | 30.4602 | 29.0258 | 10.3081 | px |
| aspect_ratio | lacuna shape | 3.3986 | 2.5391 | 2.7345 | unitless |
| eccentricity | lacuna shape | 0.8898 | 0.9192 | 0.1041 | unitless |
| solidity | lacuna shape | 0.7326 | 0.7575 | 0.0897 | unitless |
| roots_count | headline | 11.9444 | 11.5 | 6.3196 | count |
| ring_length_r30_px | headline | 497.667 | 474 | 264.072 | px |
| ring_length_r60_px | reported | 1000.28 | 1082.5 | 637.37 | px |
| owned_length_px | ownership-dependent | 1841.84 | 1849.98 | 1676.05 | px |
| edge_count | ownership-dependent | 121.667 | 118.5 | 112.207 | count |
| mean_edge_length_px | ownership-dependent | 14.6305 | 15.1613 | 4.1663 | px |
| perimeter_px | normalised | 283.011 | 249.797 | 150.102 | px |
| ring_area_r30_px2 | normalised | 8663 | 8857 | 3446.5 | px^2 |
| ring_area_r60_px2 | normalised | 18309.7 | 18341.5 | 6863.71 | px^2 |
| in_frame_fraction_r30 | normalised | 1 | 1 | 0 | unitless |
| in_frame_fraction_r60 | normalised | 0.996 | 1 | 0.0172 | unitless |
| ring_density_r30 | normalised | 0.0572616 | 0.0552114 | 0.0245128 | px^-1 |
| ring_density_r60 | normalised | 0.0528613 | 0.0497017 | 0.0266733 | px^-1 |
| roots_per_100px_perimeter | normalised | 4.4983 | 4.7642 | 2.2701 | per 100 px |
| ring_attached_length_r30_px | network v2 | 467.556 | 453.5 | 261.538 | px |
| ring_attached_length_r60_px | network v2 | 921.333 | 975.5 | 636.423 | px |
| ring_length_w_r30_px | network v2 | 568.878 | 547.86 | 302.487 | px |
| ring_length_w_r60_px | network v2 | 1140.12 | 1220.15 | 730.153 | px |
| sholl_crossings_r10 | network v2 | 11.2222 | 10.5 | 6.2737 | count |
| sholl_crossings_r20 | network v2 | 11 | 12.5 | 6.7997 | count |
| sholl_crossings_r30 | network v2 | 13.5 | 12 | 8.5078 | count |
| loop_count | loops, report only | 18.1111 | 8 | 23.5694 | count |
| loop_edge_share | loops, report only | 0.3567 | 0.3115 | 0.2591 | share |
| ring_width_mean_r30_px | width | 7.0933 | 7.1239 | 0.4159 | px |

Per field
---------

| measure | value | units |
|---|---|---|
| analysed_area_px2 | 1.00639e+06 | px^2 |
| total_skeleton_length_px | 38673 | px |
| canalicular_length_density_per_px | 0.0384274 | px^-1 |
| junction_count | 1399 | count |
| junction_density_per_px2 | 0.00139012 | px^-2 |
| lacuna_count | 21 | count |

| | | |
|---|---|---|
| lacunae_per_px2 | 2.087e-05 | px^-2 |
| skeleton_component_count | 144 | count |
| mean_component_length_px | 268.562 | px |
| median_component_length_px | 12 | px |
| field_length_density_w_per_px | 0.0435555 | px^-1 |
| field_density_without_flagged_per_px | 0.0430498 | px^-1 |
| field_density_in_roi_per_px | | px^-1 |
| unattached_length_px | 80 | px |
| unattached_component_count | 6 | count |
| canalicular_length_density_attached_per_px | 0.0384274 | px^-1 |
| field_length_density_with_unattached_per_px | 0.0385069 | px^-1 |
| loop_count | 472 | count |
| loop_edge_share | 0.5902 | share |
| edge_count | 2745 | count |

Canalicular width, the new measure

| measure | value | units |
|---|---|---|
| width_mean_px | 7.6596 | px |
| width_median_px | 8 | px |
| width_p10_px | 4 | px |
| width_p90_px | 10.198 | px |
| width_pixels_used | 36683 | count |

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

| lacuna_id | area_px2 | major_axis_length_px | minor_axis_length_px | aspect_ratio | eccentricity | solidity | orientation_rad | centroid_row_px | centroid_col_px | on_border |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 3185 | 293.107 | 22.7028 | 12.9106 | 0.997 | 0.5399 | -1.5651 | 143.896 | 232.375 | False |
| 2 | 2077 | 144.059 | 25.2241 | 5.7112 | 0.9846 | 0.6281 | 1.3769 | 151.947 | 448.976 | False |
| 3 | 4256 | 158.774 | 41.7686 | 3.8013 | 0.9648 | 0.6782 | -1.4497 | 152.645 | 678.274 | False |
| 4 | 1429 | 52.067 | 37.8414 | 1.3759 | 0.6869 | 0.8279 | -0.6051 | 154.502 | 593.396 | False |
| 5 | 1593 | 70.4006 | 31.8422 | 2.2109 | 0.8919 | 0.7622 | 1.3948 | 164.237 | 538.472 | False |
| 6 | 453 | 57.8328 | 13.3321 | 4.3379 | 0.9731 | 0.6292 | 1.5169 | 161.32 | 830.574 | False |
| 7 | 922 | 97.7946 | 15.9727 | 6.1226 | 0.9866 | 0.6272 | 1.3884 | 181.894 | 939.311 | False |
| 8 | 3025 | 100.943 | 44.409 | 2.273 | 0.898 | 0.65 | -1.4641 | 211.119 | 126.158 | False |
| 9 | 2144 | 87.4647 | 34.0354 | 2.5698 | 0.9212 | 0.7528 | -1.2557 | 226.871 | 777.66 | False |
| 10 | 2841 | 103.213 | 37.4578 | 2.7554 | 0.9318 | 0.8351 | -1.3658 | 276.205 | 420.61 | False |
| 11 | 497 | 30.0795 | 23.8394 | 1.2618 | 0.6098 | 0.7827 | -0.6524 | 264.441 | 629.968 | False |
| 12 | 3125 | 68.6713 | 60.8024 | 1.1294 | 0.4648 | 0.909 | 0.622 | 332.681 | 995.111 | True |
| 13 | 3954 | 114.494 | 48.2385 | 2.3735 | 0.9069 | 0.7517 | -1.4049 | 393.248 | 55.5005 | True |
| 14 | 3573 | 94.8523 | 52.0807 | 1.8213 | 0.8358 | 0.8284 | -1.4851 | 395.903 | 280.417 | False |
| 15 | 1409 | 76.4593 | 28.6406 | 2.6696 | 0.9272 | 0.7683 | -1.5183 | 392.216 | 831.581 | False |
| 16 | 2226 | 76.9066 | 39.7827 | 1.9332 | 0.8558 | 0.7987 | -1.558 | 506.567 | 185.868 | False |
| 17 | 1503 | 75.7312 | 26.0285 | 2.9095 | 0.9391 | 0.8628 | -1.3051 | 568.045 | 702.943 | False |
| 18 | 1123 | 61.8357 | 30.8107 | 2.007 | 0.867 | 0.6785 | -1.3397 | 588.985 | 20.1576 | True |
| 19 | 628 | 38.8866 | 23.9678 | 1.6225 | 0.7875 | 0.7235 | -1.4785 | 619.033 | 332.451 | False |
| 20 | 1563 | 73.771 | 29.411 | 2.5083 | 0.9171 | 0.717 | -1.5252 | 651.639 | 225.151 | False |
| 21 | 659 | 47.4878 | 19.9461 | 2.3808 | 0.9075 | 0.7744 | 1.5081 | 843.693 | 521.331 | False |

Network per lacuna

| lacuna_id | roots_count | ring_length_r30_px | ring_length_r60_px | owned_length_px | edge_count | mean_edge_length_px | ring_width_mean_r30_px |
|---|---|---|---|---|---|---|---|
| 1 | 19 | 668 | 1102 | 1659.8 | 122 | 13.6049 | 6.5894 |
| 2 | 8 | 344 | 557 | 549.333 | 36 | 15.2593 | 6.6102 |
| 3 | 11 | 439 | 506 | 553.76 | 44 | 12.5854 | 7.2488 |
| 4 | 4 | 181 | 210 | 208.782 | 17 | 12.2813 | 7.1239 |
| 5 | 5 | 188 | 319 | 271.238 | 15 | 18.0825 | 7.012 |
| 6 | 5 | 207 | 230 | 92.1127 | 5 | 18.4225 | 7.5208 |
| 7 | 12 | 293 | 434 | 336.764 | 19 | 17.7244 | 6.5092 |
| 8 | 23 | 840 | 1399 | 1919.47 | 115 | 16.6911 | 7.2004 |
| 9 | 19 | 722 | 1221 | 1780.49 | 122 | 14.5942 | 7.0453 |
| 10 | 14 | 754 | 1743 | 2169.29 | 139 | 15.6064 | 7.206 |
| 11 | 9 | 449 | 1046 | 3143.42 | 192 | 16.372 | 7.8128 |
| 12 | 13 | 358 | 847 | 1304.12 | 86 | 15.1641 | 7.5825 |
| 13 | 17 | 607 | 1362 | 1370.58 | 94 | 14.5807 | 7.3359 |
| 14 | 17 | 874 | 2140 | 4594.31 | 305 | 15.0633 | 7.846 |
| 15 | 12 | 409 | 1137 | 2198.56 | 126 | 17.4489 | 6.4577 |
| 16 | 20 | 783 | 1870 | 2108.23 | 171 | 12.3288 | 7.3483 |
| 17 | 9 | 506 | 1063 | 2047.17 | 113 | 18.1165 | 6.8757 |
| 18 | 12 | 482 | 1240 | 2316.1 | 155 | 14.9426 | 6.8827 |
| 19 | 11 | 499 | 1252 | 6152.76 | 413 | 14.8977 | 6.8692 |
| 20 | 17 | 802 | 1761 | 3367.6 | 236 | 14.2695 | 7.3103 |
| 21 | 0 | 0 | 15 | 0 | 0 | 0 | |

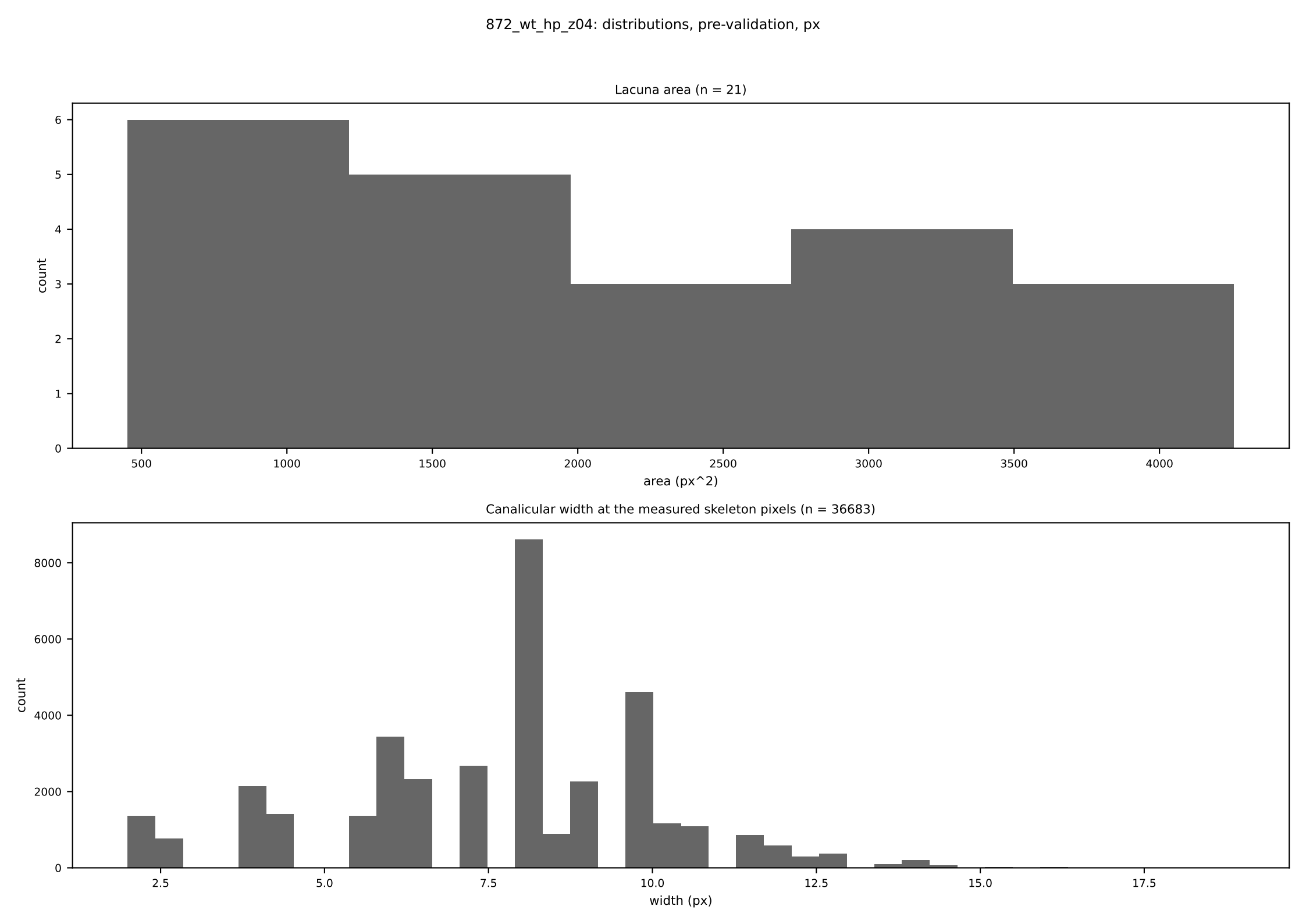
Normalised per lacuna

| lacuna_id | perimeter_px | ring_area_r30_px2 | ring_area_r60_px2 | in_frame_fraction_r30 | in_frame_fraction_r60 | ring_density_r30 | ring_density_r60 | roots_per_100px_perimeter |
|---|---|---|---|---|---|---|---|---|
| 1 | 728.713 | 18655 | 35000 | 1 | 1 | 0.0358081 | 0.0314857 | 2.6073 |
| 2 | 364.67 | 8953 | 15872 | 1 | 1 | 0.0384229 | 0.0350932 | 2.1938 |
| 3 | 505.132 | 11835 | 18345 | 1 | 1 | 0.0370934 | 0.0275825 | 2.1776 |
| 4 | 197.238 | 3616 | 6552 | 1 | 1 | 0.0500553 | 0.0320513 | 2.028 |
| 5 | 234.723 | 4437 | 8107 | 1 | 1 | 0.042371 | 0.0393487 | 2.1302 |
| 6 | 134.361 | 5080 | 7961 | 1 | 1 | 0.040748 | 0.0288908 | 3.7213 |
| 7 | 237.723 | 8825 | 18246 | 1 | 0.9271 | 0.0332011 | 0.023786 | 5.0479 |
| 8 | 425.345 | 10604 | 20758 | 1 | 1 | 0.0792154 | 0.0673957 | 5.4074 |
| 9 | 273.321 | 9051 | 16323 | 1 | 1 | 0.0797702 | 0.0748024 | 6.9515 |
| 10 | 288.978 | 10581 | 24146 | 1 | 1 | 0.0712598 | 0.0721859 | 4.8447 |
| 11 | 117.054 | 5613 | 13620 | 1 | 1 | 0.0799929 | 0.0767988 | 7.6888 |
| 12 | 236.551 | 6707 | 15609 | 0.711 | 0.6585 | 0.0533771 | 0.0542636 | 5.4956 |
| 13 | 432.996 | 10444 | 22425 | 0.8508 | 0.7745 | 0.0581195 | 0.0607358 | 3.9261 |
| 14 | 315.191 | 11229 | 25289 | 1 | 1 | 0.0778342 | 0.0846218 | 5.3936 |
| 15 | 256.208 | 8359 | 21924 | 1 | 1 | 0.0489293 | 0.051861 | 4.6837 |
| 16 | 273.764 | 9314 | 21493 | 1 | 1 | 0.084067 | 0.0870051 | 7.3055 |
| 17 | 204.445 | 8382 | 22359 | 1 | 1 | 0.0603675 | 0.0475424 | 4.4022 |
| 18 | 216.752 | 5820 | 13951 | 0.7153 | 0.655 | 0.0828179 | 0.0888825 | 5.5363 |
| 19 | 148.61 | 6134 | 15204 | 1 | 1 | 0.0813499 | 0.0823467 | 7.4019 |
| 20 | 243.386 | 8889 | 20037 | 1 | 1 | 0.0902239 | 0.0878874 | 6.9848 |
| 21 | 145.338 | 6377 | 18338 | 1 | 1 | 0 | 0.00081797 | 0 |

Network v2 per lacuna

| lacuna_id | ring_attached_length_r30_px | ring_attached_length_r60_px | ring_length_w_r30_px | ring_length_w_r60_px | sholl_crossings_r10 | sholl_crossings_r20 | sholl_crossings_r30 |
|---|---|---|---|---|---|---|---|
| 1 | 617 | 1029 | 759.014 | 1243.99 | 20 | 13 | 11 |
| 2 | 259 | 533 | 388.037 | 639.76 | 6 | 4 | 10 |
| 3 | 430 | 501 | 509.546 | 588.073 | 11 | 8 | 9 |
| 4 | 181 | 208 | 206.711 | 238.439 | 3 | 3 | 4 |
| 5 | 162 | 225 | 216.095 | 361.593 | 5 | 2 | 5 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 6 | 188 | 216 | 237.752 | 262.995 | 6 | 6 | 3 |
| 7 | 261 | 286 | 328.35 | 486.061 | 7 | 6 | 7 |
| 8 | 809 | 1173 | 968.737 | 1605.09 | 22 | 15 | 16 |
| 9 | 645 | 1168 | 830.654 | 1403.41 | 16 | 14 | 21 |
| 10 | 732 | 1683 | 863.654 | 1997.43 | 18 | 19 | 22 |
| 11 | 448 | 1011 | 530.588 | 1196.31 | 9 | 13 | 11 |
| 12 | 308 | 675 | 399.635 | 964.708 | 7 | 12 | 12 |
| 13 | 564 | 1127 | 706.683 | 1562.71 | 14 | 14 | 14 |
| 14 | 854 | 2137 | 998.767 | 2464.55 | 19 | 25 | 31 |
| 15 | 342 | 940 | 458.605 | 1271.38 | 9 | 12 | 13 |
| 16 | 779 | 1853 | 884.441 | 2140.3 | 14 | 17 | 27 |
| 17 | 459 | 740 | 574.132 | 1183.22 | 11 | 8 | 15 |
| 18 | 475 | 1239 | 544.676 | 1418.12 | 12 | 12 | 14 |
| 19 | 466 | 1147 | 565.132 | 1416.15 | 10 | 14 | 20 |
| 20 | 784 | 1734 | 919.583 | 2005.28 | 16 | 19 | 18 |
| 21 | 0 | 0 | 0 | 18.0711 | 0 | 0 | 0 |


872_wt_hp_z04: distributions, pre-validation, px
Lacuna area (n = 21)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 36683)
count
width (px)